

Task 5 — The Discovery Phase (EDA)

Data Analyst · Phase 1 Industry Immersion · Study Guide

DATA ANALYST

PlaceMux · Phase 1 · Study Guide

DIFFICULTY

Foundational · Basic

Level 5 of 20 · ramp 24%

1 · What this is & why it matters

Run Exploratory Data Analysis (EDA): use stats and visuals to discover the structure, patterns and surprises in the data.

EDA is where genuine insight starts and where you catch problems no metric would reveal. It shapes every question you ask next.

2 · Learning objectives

By the end of this guide and task you will be able to:

- Deliver: An EDA notebook with annotated charts.
- Deliver: A shortlist of candidate insights and follow-up questions.
- Explain the key concepts behind the task in your own words.
- Choose between alternative ways to execute it, and justify your choice.
- Avoid the common pitfalls and know how your work is scored.

3 · Prerequisites

- A reproducible Python data environment (pandas, numpy, matplotlib).
- Comfort reading data into a DataFrame and writing basic SQL.
- The habit of validating data before analysing it.

4 · Key concepts (understand these first)

Univariate analysis

one variable at a time (distribution, central tendency, spread).

Bivariate/multivariate

relationships between variables (scatter, grouped stats).

Distribution shape

normal, skewed, bimodal -- each implies different methods.

Segmentation

slicing by category to see where behaviour differs.

Hypothesis generation

EDA produces questions to test, not final answers.

5 · Recommended stack & tools

- pandas groupby
- matplotlib / seaborn
- Jupyter

6 · The build pipeline (follow in order)

This is the flow from nothing to demoable. Each step builds on the last; the final steps verify and prove the work.

1. Plot the distribution of every key numeric variable.
2. Compare key variables across important categories (group-bys).
3. Look at relationships between pairs of variables with scatter plots.
4. Identify and investigate the most surprising patterns.
5. Note candidate insights and the questions they raise.
6. Summarise findings with the 3-4 most decision-relevant charts.

7 · Other ways to execute (alternative approaches)

There is rarely one right path. Consider these alternatives and pick deliberately for your context:

- Manual EDA vs an automated EDA report -- automated is fast but you learn less.
- Static charts vs interactive (Plotly) for richer exploration.

8 · Brainstorming & deeper thinking

Sit with these questions — they separate someone who completed the task from someone who understood it:

- Which finding, if true, would change a real decision?
- Are surprising patterns real or artefacts of cleaning?
- What's the single chart that best tells today's story?

9 · Definition of Done

Done when all are true and demoable

- An EDA notebook with annotated charts.
- A shortlist of candidate insights and follow-up questions.
- Demonstrable live on real data.

10 · How your work is evaluated (out of 100)

Scoring parameter	Marks
Core deliverable — An EDA notebook with annotated charts.	25
Core deliverable — A shortlist of candidate insights and follow-up questions.	25
Real-data quality & correctness (real inputs at realistic scale, not a toy/happy-path)	20
Live verification & evidence (demonstrated live; real output, not claims)	15
Dependency, failure & edge-case handling (errors handled; hand-offs honoured)	15
TOTAL	100

11 · Pitfalls to avoid

Worry if any of these are true of your work

- Making 50 charts and no point.
- Confusing correlation with cause this early.
- Polishing visuals before understanding them.

12 · Where this fits & go deeper

Over Phase 1 you grow from loading a CSV to shipping a live, decision-grade analytics product. You turn raw data into numbers people trust and decisions people make.

Go deeper (optional further study):

- Statistics for analysts (distributions, tests, intervals)
- SQL window functions and query optimisation
- Storytelling with data and dashboard design
- Time-series and segmentation techniques